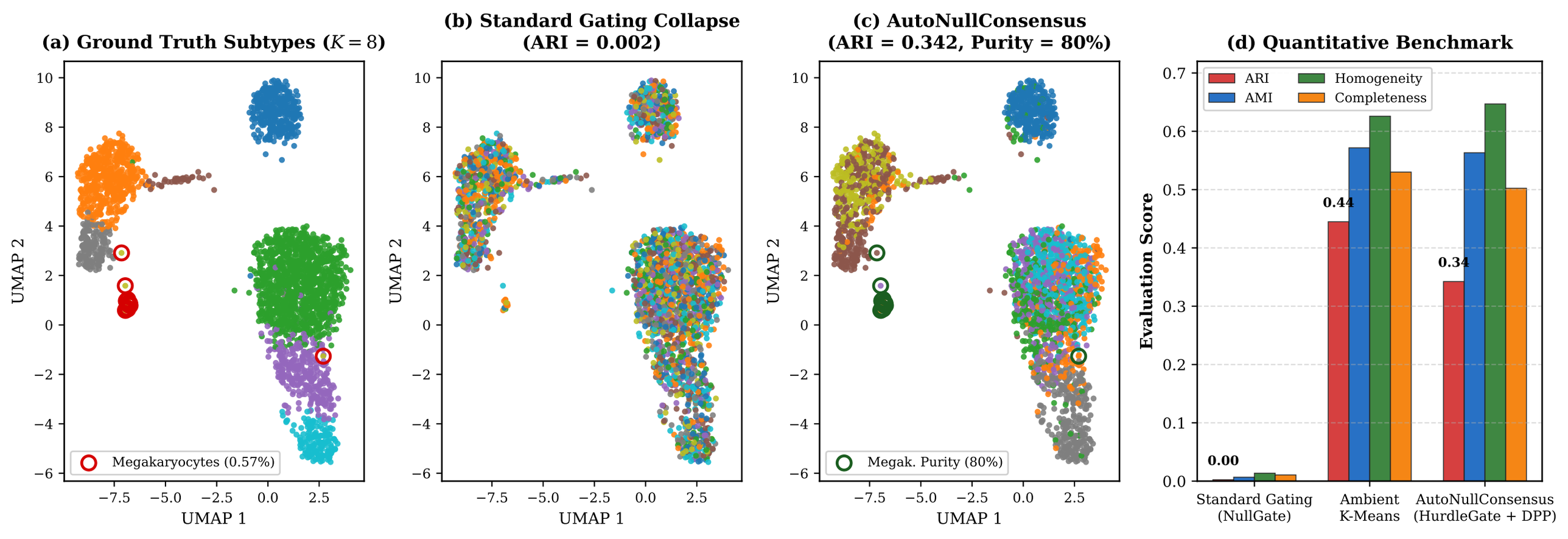

(a) Ground Truth Subtypes ($K = 8$)
(b) Standard Gating Collapse
(ARI = 0.002)
(c) AutoNullConsensus
(ARI = 0.342, Purity = 80%)
(d) Quantitative Benchmark
UMAP 1
UMAP 2
Megakaryocytes (0.57%)
Megak. Purity (80%)
ARI
AMI
Homogeneity
Completeness
Evaluation Score
0.00
0.44
0.34
Standard Gating
(NullGate)
Ambient
K-Means
AutoNullConsensus
(HurdleGate + DPP)